

Jafsal M A

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PROFESSIONAL SUMMARY

Senior Data Engineer and Technical Lead with 7+ years building large-scale web crawling systems and Python ETL pipelines, delivering production-grade ingestion, transformation, and automated delivery at 100Ks-to-millions-of-records scale. Experienced leading data operations teams end-to-end - from crawling through client-format delivery (SAP XML, E4H XML) - and currently moving into agentic AI, building automation agents on the Hermes framework.

WORK EXPERIENCE

PST.AG

Feb 2024 - Present

Python Developer / Team Lead - Data Engineering & Web Scraping

Weimar, Germany (Remote)

- Led design and delivery of Python backend tooling for a scraping ecosystem of 100+ production projects, improving ingestion throughput and downstream data readiness for analytics
- Led the Data Operations team for one year, owning the full ETL project lifecycle - crawling, transformation, validation, and automated delivery - with outputs delivered to clients in SAP XML, E4H XML, and other structured formats
- Built and scaled ingestion pipelines across distributed storage (ArangoDB, Elasticsearch) to process high-volume multi-source feeds with fault tolerance and schema evolution support
- Designed and operated ETL workflows using Apache NiFi for ingestion, transformation, routing, and automated validation of heterogeneous data streams
- Orchestrated ETL workflows with Apache Airflow
- Implemented automated validation and scenario-based verification (BDD) to detect data anomalies earlier and reduce incorrect downstream records
- Optimized the automated file-delivery script used to ship customer deliverables on each release, improving delivery reliability and reducing manual intervention
- Translated client and stakeholder requirements into scoped technical work, contributing to project planning, architecture documentation, and technical specifications
- Completed onboarding for the company's move into agentic AI and joined the team building automation agents on the Hermes framework, integrating MCP and Matternmost for operational communication
- Led a team of 4 engineers, conducting code reviews, mentoring, and driving internal tooling and performance optimization POCs
- Maintained containerized services on Docker, Scrapy, and Linux infrastructure, integrating releases into CI/CD pipelines for versioned rollouts and monitoring
- Participated in Agile ceremonies and delivery tracking via Jira

Shopalyst Technologies

Apr 2021 - Jan 2024

Software Engineer II - Data Operations and Engineering

Trivandrum, Kerala, India

- Architected Python and Golang backend systems that processed millions of records daily, improving pipeline throughput and operational reliability
- Reverse-engineered Android applications, including SSL-pinned apps, to discover internal endpoints and integrate RESTful APIs into structured pipelines, using BrowserStack and LambdaTest for cross-device testing and validation
- Designed and enforced automated verification pipelines and data quality checks to ensure correctness across extracted datasets
- Operated distributed storage and messaging to support high-throughput workflows, using Elasticsearch, Cassandra, and RabbitMQ
- Used Apache Spark (PySpark) for distributed data processing in the data pipelines
- Automated operational workflows with Bash, Cron, and Jenkins, reducing manual failures and deployment errors
- Mentored junior engineers and maintained system stability through proactive monitoring, debugging, and incident resolution

Tata Consultancy Services

Dec 2020 - Mar 2021

IoT Edge Developer

Kochi, Kerala, India

- Architected modular C++ services for Azure IoT Edge to enable reliable, low-latency processing across distributed edge environments
- Delivered Dockerized services and automated builds using Azure DevOps and Git to support continuous integration and deployment
- Implemented protocol-heavy solutions with strict validation requirements to ensure compliance and system integrity in regulated environments

Tata Consultancy Services

Dec 2018 - Nov 2020

System Engineer

Chennai, Tamil Nadu, India

- Developed Python ETL crawlers using Scrapy and BeautifulSoup to extract, clean, and transform e-commerce data supporting downstream analytics and ML feature pipelines
- Built parsing and feature-extraction logic that improved data completeness and reduced downstream cleaning effort, using regular expressions and pandas
- Documented crawling workflows, contributed to code review processes, and supported knowledge transfer to maintainable scraping codebases

PROJECTS

Rapid Prototyping & AI-Assisted Tooling (Vibe Coding)

Built a full-stack food ordering web app (dev.theplato.in), owning UI, backend logic, data persistence, and deployment

Firebase Studio, Antigravity

EDUCATION

Bachelor of Technology, Computer Science and Engineering [NSS College of Engineering, University of Calicut](#)

Aug 2014 - Jul 2018

Palakkad, Kerala, India

PUBLICATIONS

"Different Approaches To Crack Monitoring of Buildings Using IoT," [IJRASET Publications](#), Aug 2018

SKILLS

Data Ingestion & ETL: ETL/ELT workflows, Apache NiFi, Apache Airflow, large-scale ingestion pipelines, workflow orchestration, multi-source data acquisition, API reverse-engineering, Scrapy spiders, Scrapyd, web crawling, anti-bot countermeasures, headless-browser evasion, proxy tunnelling

Web & Parsing: Scrapy, BeautifulSoup, Selenium, Playwright, curl_cffi, HTML, JavaScript, regular expressions, pandas, RESTful APIs, GraphQL APIs

Agentic AI: Hermes Agent framework, MCP (Model Context Protocol), automation agents

Languages: Python (Advanced), SQL, Golang, Java, C++, Bash

Big Data & Storage: Apache Spark (PySpark), distributed storage, partitioning, indexing, schema evolution, Elasticsearch, Cassandra, ArangoDB, PostgreSQL, MySQL, MongoDB, RabbitMQ, time-series databases

Infrastructure & DataOps: Docker, Linux, CI/CD (GitLab, Jenkins, Azure DevOps), Git, Cron, Jira

Cloud & Platforms: Azure, Azure IoT Hub, Azure IoT Edge, Firebase

Actively Learning: Databricks (Delta Lake, Unity Catalog), Medallion architecture (bronze/silver/gold), Azure Data Factory, Kubernetes

Data Governance: Data lineage, data contracts, scenario-based validation, QA, PII/compliance, logging & fault analysis

Methodologies: Agile/Scrum, BDD, system design, code reviews, architecture documentation, training sessions, project planning & execution

Spoken Languages: English (Fluent), Malayalam (Native), Tamil (Beginner)